

Ivan Evdokimov, Ph.D.

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Professional Summary

Full Stack Developer with 4+ years of experience designing, building, and operating scalable, resilient, and secure applications using Java, Python, and Go. Expertise in cloud-native engineering on AWS, CI/CD automation, and end-to-end full-stack development (backend APIs, frontend UIs, and databases). Proven track record delivering production-grade systems with automated testing, deployment pipelines, and operational excellence in hybrid multi-cloud ecosystems.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK Nov 2025 — Present

- Designed and deployed full-stack applications using Java (Maven framework), Python, and GitHub Actions/Vercel CI/CD pipelines, enabling automated build, test, and deployment with 90%+ code generation rate and weekly release cycles in a hybrid cloud environment.
- Developed backend services with integrated LLM support for asset examination reporting, ensuring scalable, resilient application design and adherence to risk management frameworks and security policies.
- Built automated testing and QA frameworks for AI-generated code, implementing validation pipelines that reduced manual review time by 70% while maintaining code quality, operational reliability, and compliance with enterprise standards.
- Architected end-to-end MLOps infrastructure for a multi-tenant SaaS platform, accelerating feature delivery through automated workflows, infrastructure-as-code principles, and self-service capabilities for stakeholders.

Software Engineer – GenAI, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Engineered cloud-native applications on AWS using Lambda, Step Functions, and EFS, deploying serverless services for automated metadata classification and disclosure risk assessment with 95%+ accuracy, adhering to technology standards and data regulations.
- Developed and maintained CI/CD pipelines for model retraining and deployment automation, reducing manual validation workload by 50% and enabling rapid, efficient delivery of new features while ensuring service reliability and performance.
- Optimized performance-critical algorithms by migrating from R to C++, achieving 11x speed improvement for real-time processing of 500k+ record datasets, demonstrating backend engineering excellence and scalability.

Research Officer & ML Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative models from MATLAB to C++, reducing runtime by 8x through performance engineering and optimized backend computation, ensuring resilient, high-performance services.
- Taught Data Structures, C/C++, and ML to 100+ students, promoting code quality, collaboration, and Developer Manifesto principles (owning code, meaningful commit messages, clean repositories).

Projects

Chat-LLM github.com/ivan020/ChatLLM

- Backend service: Developed a Python-based self-hosted Ollama MCP server on Raspberry Pi with Twitch chat integration, demonstrating scalable backend architecture, real-time data processing, and automated operational support.

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Full-stack development: Built a Go backend API with Twitch chat integration and PostgreSQL database, paired with a JavaScript/React frontend UI for real-time portfolio visualization, demonstrating end-to-end application development from backend to frontend.
- Cloud deployment: Hosted frontend on Vercel and managed backend services, showcasing cloud-native deployment and scalable architecture.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Full-stack development: Engineered a Golang REST API with LLM integration and PostgreSQL backend, paired with a React frontend deployed on Vercel, demonstrating cloud-native engineering and modern web development.
- Operational ownership: Adhered to "you build it, you run it" principles, ensuring automated testing, deployment, and monitoring for service reliability.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Research and developed PyTorch-based frameworks for financial forecasting, publishing peer-reviewed papers and releasing open-source packages, demonstrating backend engineering and data-driven application development.
- Main author on an internationally acknowledged scientific journal publication and co-authored 3 conference papers.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Engineered C++ agent-based financial simulations to model macroeconomic dynamics, applying software engineering fundamentals to complex backend systems.